

Weather Trend Forecasting

PM Accelerator Mission

By making industry-leading tools and education available to individuals from all backgrounds, we level the playing field for future PM leaders. This is the PM Accelerator motto, as we grant aspiring and experienced PMs what they need most – Access. We introduce you to industry leaders, surround you with the right PM ecosystem, and discover the new world of AI product management skills.

Objective

The goal of this project is to examine global weather data and predict future weather patterns using machine learning algorithms. The data includes global daily weather details such as temperature, humidity, rainfall, wind speed, atmospheric pressure, and air quality indices, among others.

Basic Assesment

Data Cleaning and Preprocessing

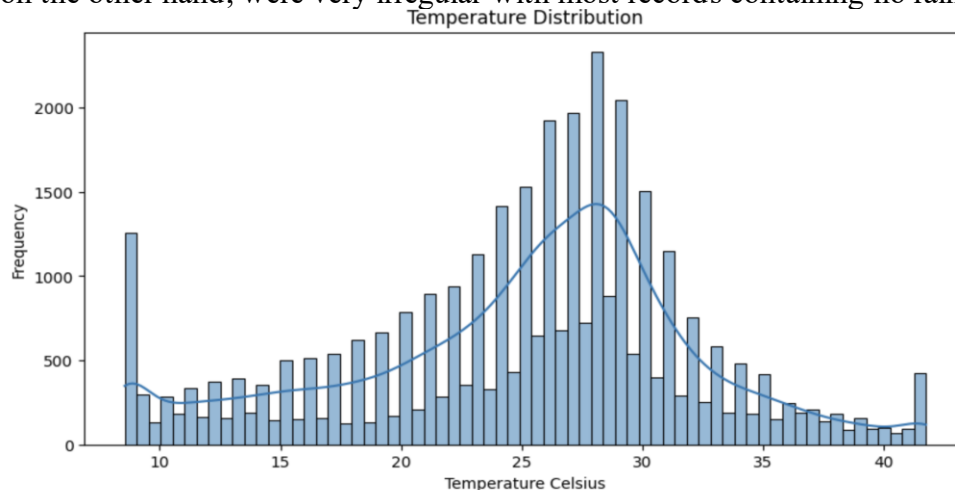
Prior to analysis, the data was prepared for quality assurance purposes.

The following actions were taken:

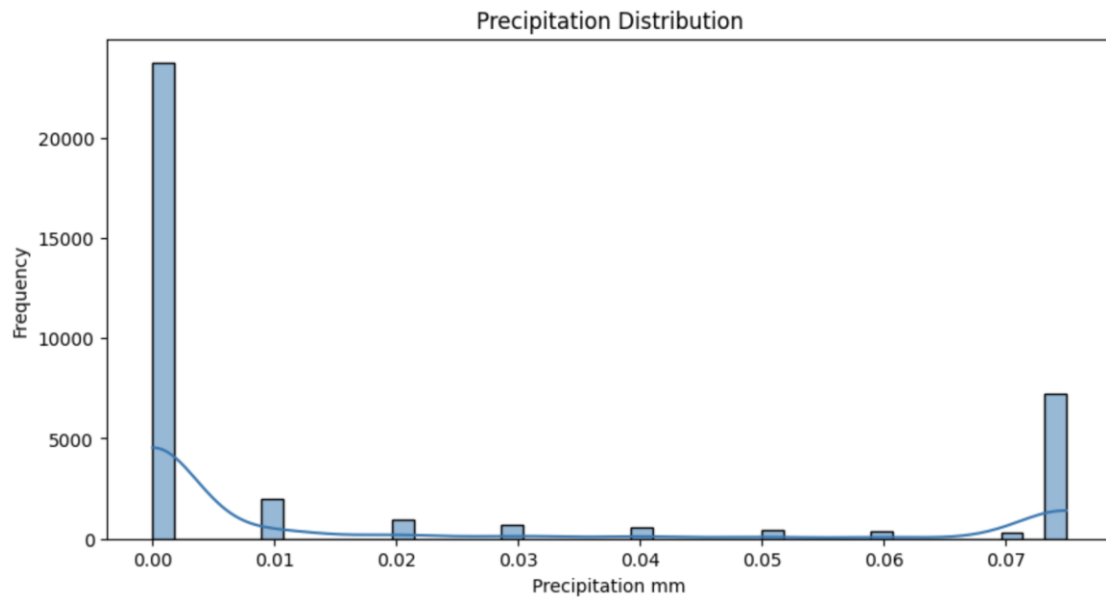
- Duplicates were dropped from the dataset.
- The last_updated field was changed to datetime format.
- Missing values for numerical fields were filled with the median.
- Missing values for categorical fields were filled with the mode.
- Outliers were handled using the IQR method.
- Extra date columns like year, month, day, and day of year were created.

Exploratory Data Analysis (EDA)

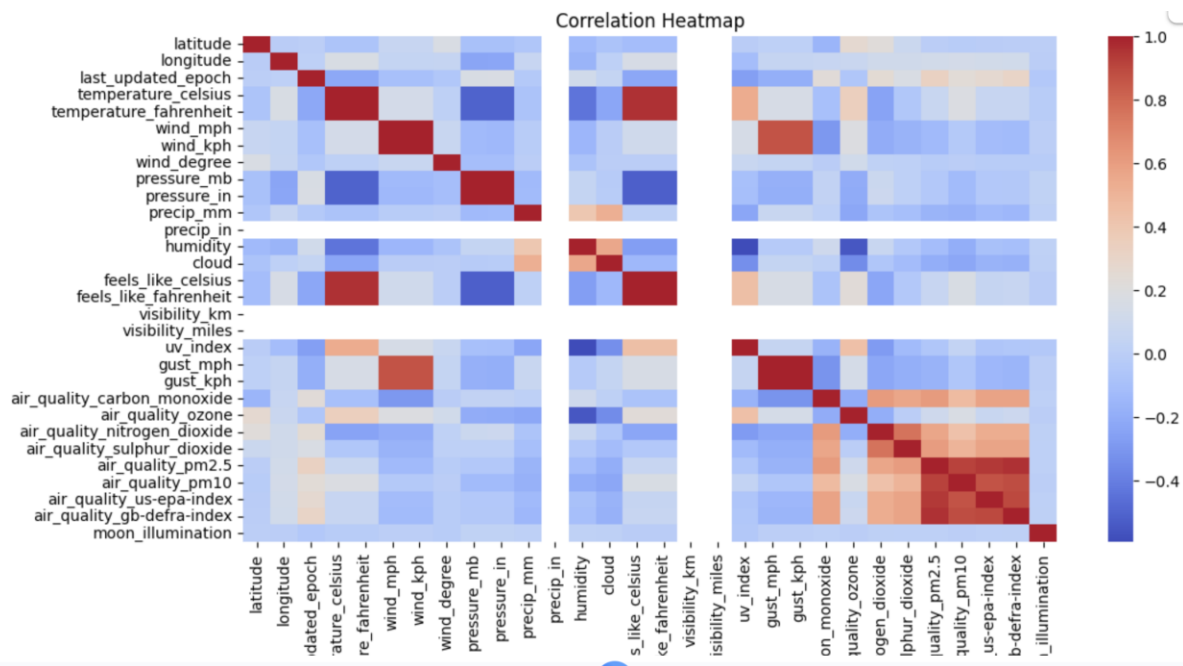
EDA was conducted to have a good insight into the data set. Temperature readings displayed high variability among regions and during various times of the year. Precipitation readings, on the other hand, were very irregular with most records containing no rain at all.



Observation: Most of the temperature recordings lie in the mid-range category; extreme temperatures happen less frequently.

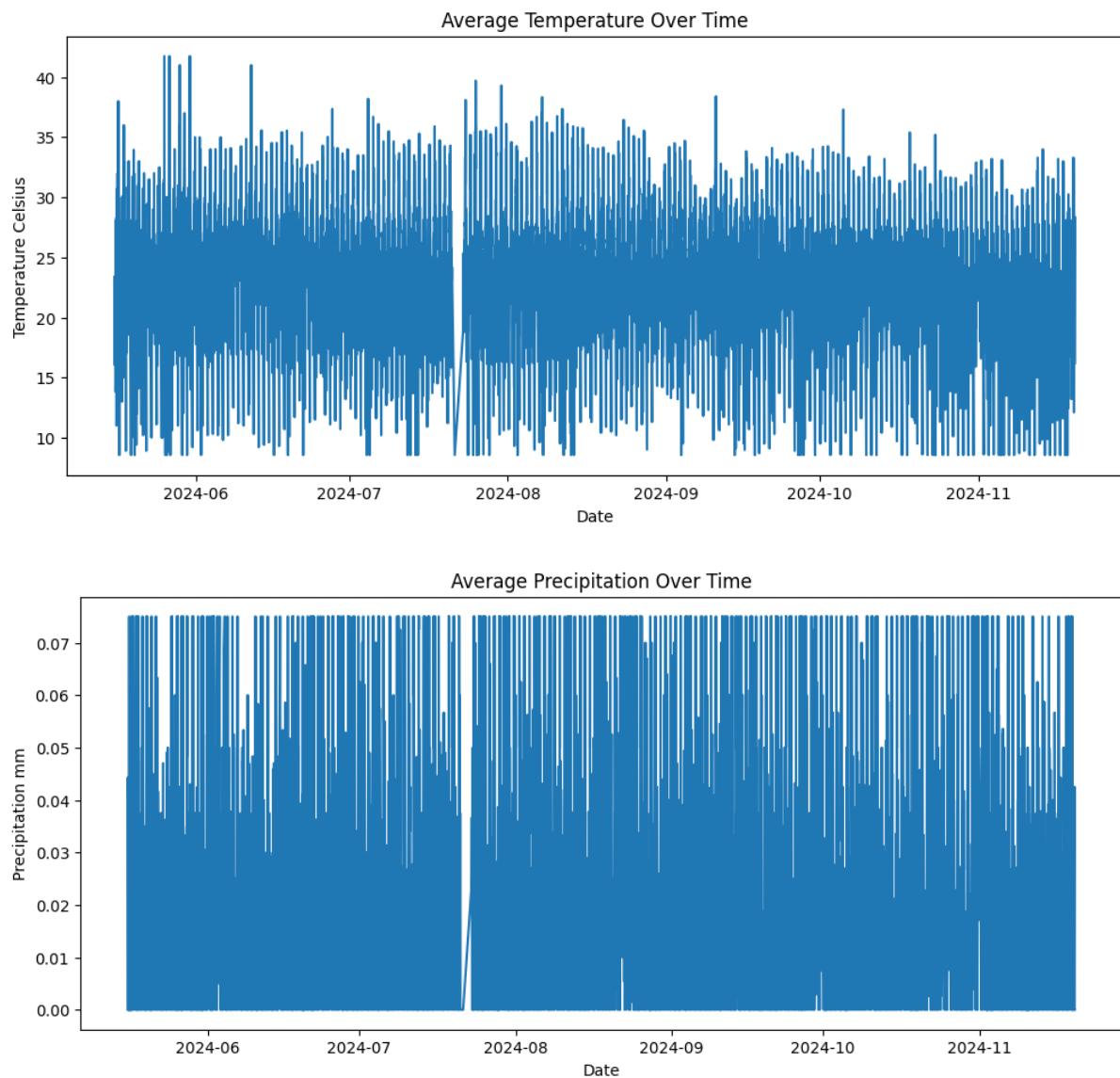


Observation: The rainfall data is very skewed because there are few days of rain and some days where there is a lot of rain.



Observation: There are various variables such as humidity, clouds, pressure, and temperature, which have significant relationships useful for predictions.

Visualizations for temperature and precipitation



These graphs offer valuable insight into long-term weather trends and seasonality.

Model Building

The models used were: Linear Regression, Random Forest Regressor, Gradient Boosting Regressor, Ensemble Model

The target variable was temperature_celsius.

Input features included: Humidity, Pressure, Wind Speed, Cloud Cover, UV Index, Precipitation, Date-related features

Random Forest Results

MAE: 3.926829508951171

RMSE: 5.0008990141871745

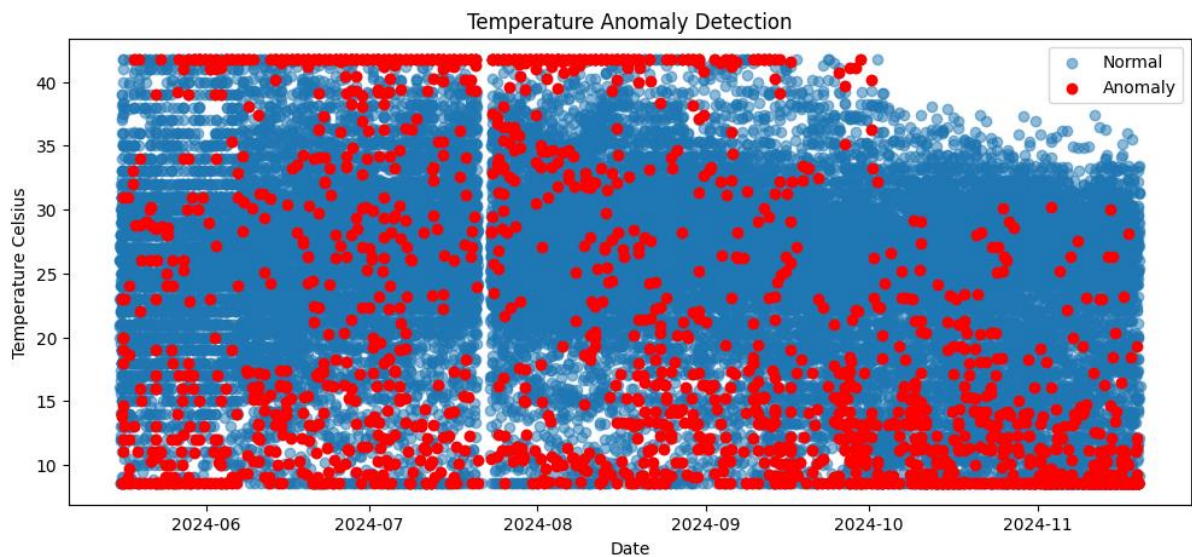
R2 Score: 0.5795144439461525

Advanced Assessment

Advanced EDA

Anomalies were detected using the Isolation Forest algorithm in order to determine unusual weather observation cases.

Number of anomalies: 1811



Observation: The highlighted dots indicate unusual weather cases which are very different from regular weather patterns.

Forecasting with Multiple Models

The prediction of future temperatures was achieved using three machine learning models that were compared for their performance. These machine learning models are the Linear Regression model, Random Forest Regressor, and Gradient Boosting Regressor. The evaluation measures considered are MAE, RMSE, and R2 score.

	Model	MAE	RMSE	R2 Score
0	Linear Regression	4.685725	5.614132	0.470068
1	Random Forest	3.926830	5.000899	0.579514
2	Gradient Boosting	4.125933	5.212984	0.543093

Ensemble Model

In order to increase prediction precision even further, an Ensemble Model was developed by using average prediction values from the Linear Regression, Random Forest, and Gradient Boosting models.

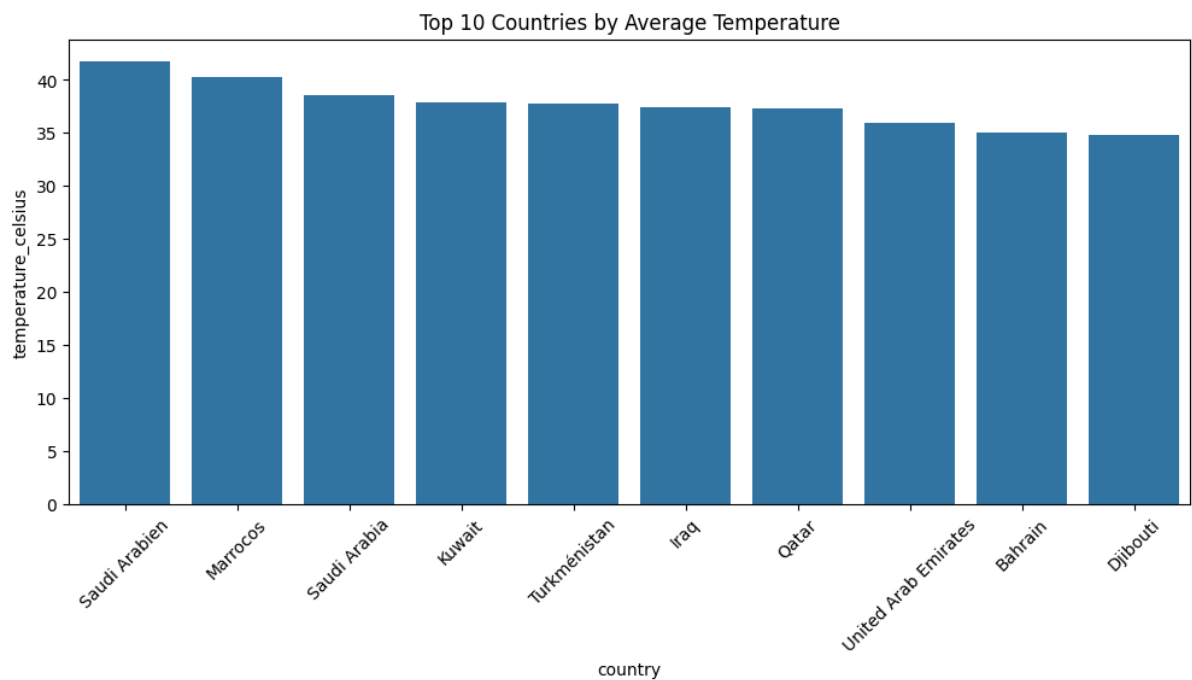
Ensemble Model Results

MAE: 4.079160311199472, RMSE: 4.967428885593164, R2 Score: 0.5851240790088034

Unique Analyses

Climate Analysis

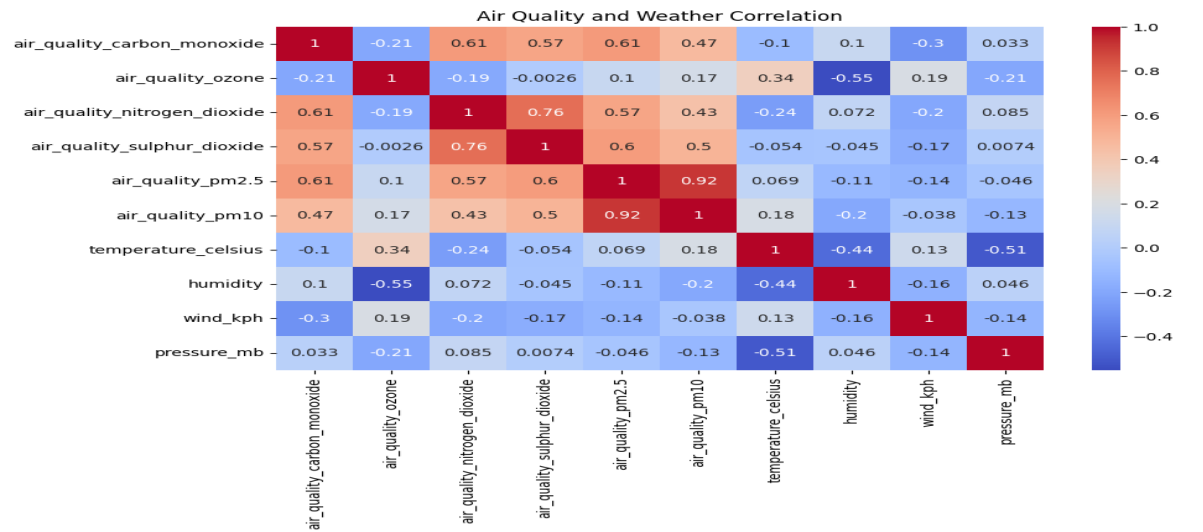
Analysis on the climate trends was carried out through observation of the average weather patterns of various countries and geographical regions.



Observation: Countries that belong to the category of deserts and arid environments had the highest averages for temperatures and lowest values for precipitation. Saudi Arabia, Kuwait, Qatar, and UAE belong to the group of places where the highest temperatures were observed, thus emphasizing the importance of geographical and climatic influences on weather.

Environmental Impact:

Analysis of the relationship between air quality indicators and the weather data was conducted.

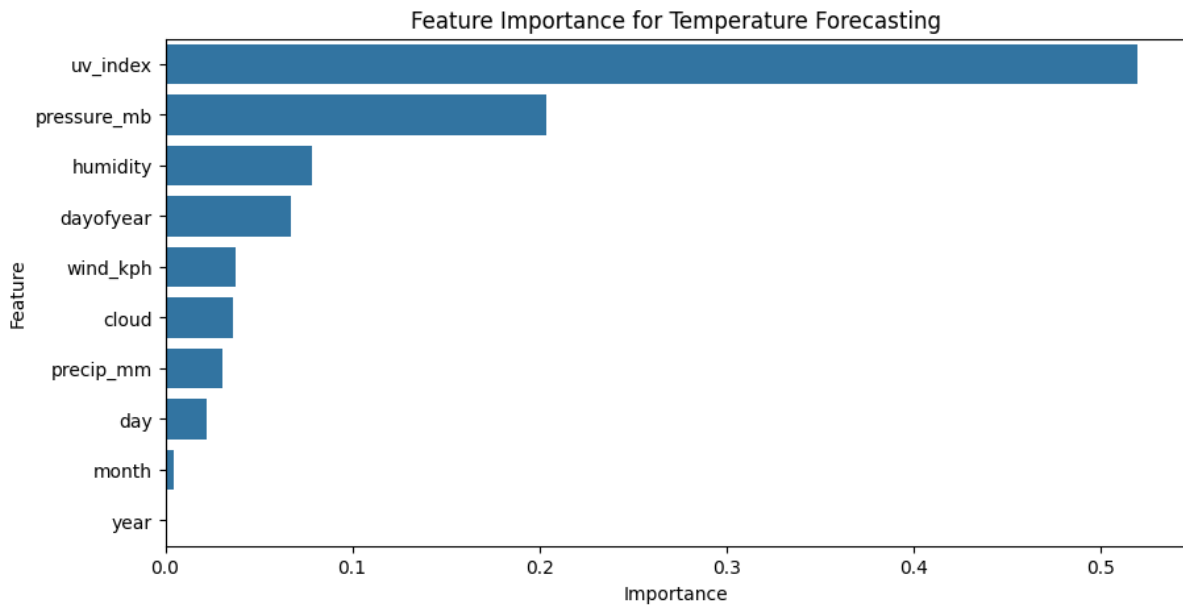


Observation: From this report, PM2.5 and PM10 demonstrate a high correlation, thus meaning that both types of pollutants tend to increase together. In addition, temperature, humidity, and pressure have some relationships with air quality indexes, which means that

weather impacts pollution.

Feature Importance:

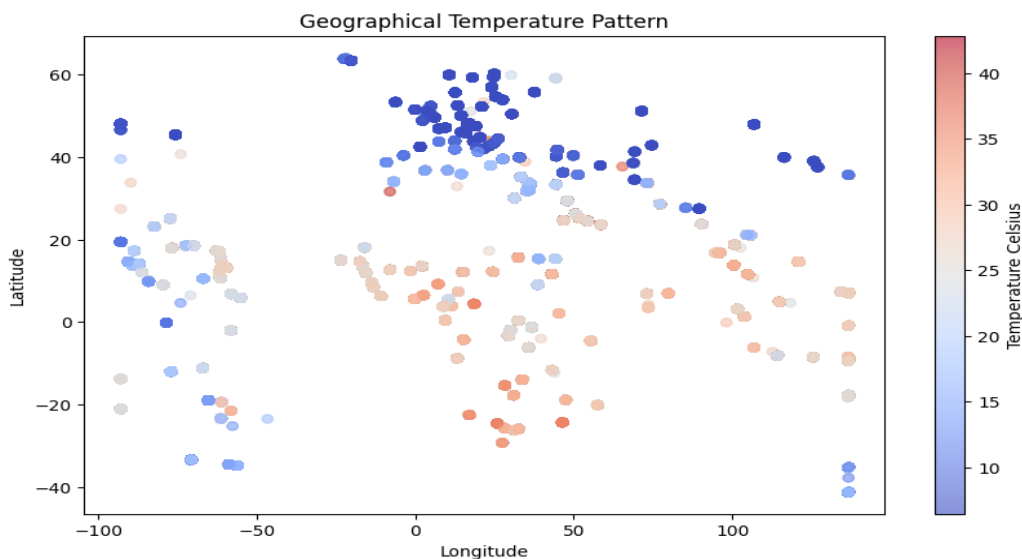
The feature importance analysis was conducted using Random Forest.



Observation: The greatest impact on predicting temperature is provided by the UV Index. Alongside with pressure and humidity, this factor helps to predict temperature in a more accurate way.

Spatial Analysis:

Spatial Analysis was done based on the longitude and latitude data to visualize temperature patterns across different regions of the world.



Observation: According to the analysis of spatial data, there is a significant difference in terms of temperature depending on geographic location. It was found out that countries with geographical coordinates close to the equator usually have higher temperatures than those

located further from the equator.

Geographical Patterns

The geographical patterns of weather have been studied by analyzing temperature, humidity, rainfall, and wind speed differences among different countries and cities.

..		country	location_name	temperature_celsius	humidity
0		Afghanistan	Kabul	24.403623	22.700483
1		Albania	Tirana	26.483816	48.342995
2		Algeria	Algiers	26.356311	54.849515
3		Andorra	Andorra La Vella	15.768841	55.971014
4		Angola	Luanda	26.170531	70.483092
5		Antigua and Barbuda	Saint John's	28.207843	77.661765
6		Argentina	Buenos Aires	12.899029	81.427184
7		Armenia	Yerevan	24.638406	32.526570
8		Australia	Canberra	10.476942	78.927184
9		Australia	Melbourne	11.600000	54.000000
10		Austria	Vienna	19.945146	58.762136
11		Azerbaijan	Baku	23.833010	54.567961
12		Bahamas	Nassau	27.356650	77.960591
13		Bahrain	Manama	34.017874	50.043478
14		Bangladesh	Dhaka	30.216505	63.330097
15		Barbados	Bridgetown	28.501942	80.262136
16		Belarus	Minsk	18.232126	62.565217
17		Belgium	'S Gravenjansdijk	16.955650	72.333333
18		Belgium	'S Gravenjansdyk	16.341667	67.583333
19		Belgium	'S Gravenstaffel	15.562500	80.500000
...		precip_mm	wind_kph		
0		0.004928	12.942995		
1		0.018575	11.932367		
2		0.005218	16.039078		
3		0.027705	9.739130		
4		0.003575	13.561353		
5		0.031936	18.984804		
6		0.007621	11.049029		
7		0.011449	9.009420		
8		0.006141	10.330825		
9		0.000000	18.000000		
10		0.012743	14.861408		
11		0.008447	21.866505		
12		0.013916	18.283005		
13		0.000000	19.703140		
14		0.026505	15.523058		
15		0.034199	22.871117		
16		0.017778	16.355556		
17		0.026186	17.833051		
18		0.016667	20.662500		
19		0.038437	17.125000		

Observation: From the geographical analysis, there are visible differences in weather patterns in different countries and cities. The warmest places like Bahrain and Bangladesh had high average temperatures. Places like Australia and Argentina were cool regions.